A WORKED FINITE-HORIZON LIKELIHOOD EXAMPLE FOR THE RANDOM-LIFETIME FROG MODEL

JOSÉ HERMENEGILDO RAMÍREZ GONZÁLEZ

Abstract. This note instantiates the notation used in the accompanying article and supplementary material for one finite labeled record. The record consists of $K = 5$ independent realizations observed through horizon $T = 4$ and generated at $\beta_0 = 0.45$. Activation times, labeled trajectories, censoring variables, particle–age indicators, age-specific counts, the three point estimators, and the exact observed-data likelihood are displayed for the same realization. Latent marks and complete potential lifetimes are shown only because the record is simulated; the statistical observation remains the labeled state–position record through time T.

1. Numerical setting

There is one particle at each site of $\mathbb{Z}$, and only the origin particle is active at calendar time 0. A particle is indexed by (r, x), where $r \in \{1, \ldots, 5\}$ denotes the realization and $x \in \mathbb{Z}$ its home site. For each label (r, x), the latent survival mark satisfies

$$\Pi_{r,x} \sim \mathrm{Beta}(1, 0.45).$$

Conditional on $\Pi_{r,x} = u$, the potential lifetime $\Xi_{r,x}$ has mass function

$$\mathbb{P}(\Xi_{r,x} = j \mid \Pi_{r,x} = u) = u^j(1 - u), \qquad j \in \mathbb{N}_0.$$

Thus $\Xi_{r,x}$ is the number of jumps that the particle would complete after activation and before death. Independently of the lifetime variables, the particle receives an i.i.d. direction sequence $\varepsilon_{r,x}(m) \in \{-1, +1\}$ with equal probabilities.

The record is generated with random seed 9. From independent uniforms U_1, U_2, the latent variables are obtained as

$$\Pi = 1 - (1 - U_1)^{1/\beta_0}, \qquad \Xi = \left\lfloor \frac{\log U_2}{\log \Pi} \right\rfloor,$$

followed by symmetric directions. The value $\beta_0 = 0.45$ is used only to generate the record; throughout the likelihood calculations, β is treated as unknown.

Remark 1.1. *The quantities $\Pi_{r,x}$, $\Xi_{r,x}$, and the unobserved future directions are latent. They are reported only to make the construction transparent. In the corresponding statistical experiment, the observed data are the particle states and positions at calendar times $0, 1, 2, 3, 4$.*

2. Notation instantiated by the record

For an activated particle (r, x), the variables used below have the following meanings.

$A_{r,x}$: The calendar time at which the particle is activated. It is dormant before $A_{r,x}$ and active at time $A_{r,x}$.

$X_{r,x}(m)$: The potential position after m completed jumps:

$$X_{r,x}(0) = x, \qquad X_{r,x}(m) = x + \sum_{i=1}^{m} \varepsilon_{r,x}(i).$$

$H_{r,x}(t)$: The state at calendar time t: dormant (sl), active and alive (al), or dead (de).

$L_{r,x}(t)$: The recorded position at time t. Before activation it equals the home site; after death it remains at the last occupied site.

$C_{r,x}(T)$: The number of transition opportunities before the horizon:

$$C_{r,x}(T) = T - A_{r,x} \quad \text{if } A_{r,x} < T, \qquad C_{r,x}(T) = 0 \quad \text{if } A_{r,x} = T.$$

$Z_{r,x}(T)$: The observed lifetime length,

$$Z_{r,x}(T) = \min\{\Xi_{r,x}, C_{r,x}(T)\},$$

namely the number of jumps observed before death or censoring.

$\Delta_{r,x}(T)$: The observed-death indicator,

$$\Delta_{r,x}(T) = \mathbf{1}_{\{\Xi_{r,x} < C_{r,x}(T)\}}.$$

The value 0 denotes right censoring at the horizon.

$Y_{r,x,j}(T)$: The indicator that the age-j transition is visible:

$$Y_{r,x,j}(T) = \mathbf{1}_{\{A_{r,x}+j<T,\ \Xi_{r,x} \geq j\}}.$$

$D_{r,x,j}(T)$: The indicator that a visible age-j transition ends in death:

$$D_{r,x,j}(T) = \mathbf{1}_{\{A_{r,x}+j<T,\ \Xi_{r,x} = j\}}.$$

$J_{r,x,j}(T)$: The visible-jump indicator,

$$J_{r,x,j}(T) = Y_{r,x,j}(T) - D_{r,x,j}(T).$$

$Z_n^{(5)}$: The nth particle in the global activation order across the five realizations. This symbol is distinct from the observed lifetime length $Z_{r,x}(T)$.

3. Labeled trajectories and activation order

Figure 1 displays the labeled trajectories of every activated particle. An open circle marks activation, a cross marks an observed death, and an open square marks a particle that remains alive when observation stops at $T = 4$.

The five realizations are summarized below.

Realization 1. Particle $(1, 0)$ is activated at time 0, completes three jumps, and is observed to die. Particle $(1, 1)$ is activated at time 1, completes two jumps, and is observed to die.

Realization 2. Particle $(2, 0)$ is activated at time 0, completes no jump, and is observed to die.

Realization 3. Particle $(3, 0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(3, -1)$ is activated at time 1, completes three observed jumps, and is right-censored. Particle $(3, -2)$ is activated at time 2, completes no jump, and is observed to die.

Realization 4. Particle $(4, 0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(4, 1)$ is activated at time 1, completes three observed jumps, and is right-censored. Particle $(4, 2)$ is activated at time 2, completes one jump, and is observed to die. Particle $(4, -1)$ is activated at time 3, completes no jump, and is observed to die. Particle $(4, 3)$ is activated at time 3, completes one observed jump, and is right-censored. Particle $(4, 4)$ is activated at time T and therefore contributes no transition of its own.

Realization 5. Particle $(5, 0)$ is activated at time 0, completes four observed jumps, and is right-censored. Particle $(5, -1)$ is activated at time 1, completes no jump, and is observed to die. Particle $(5, 1)$ is activated at time 3, completes one observed

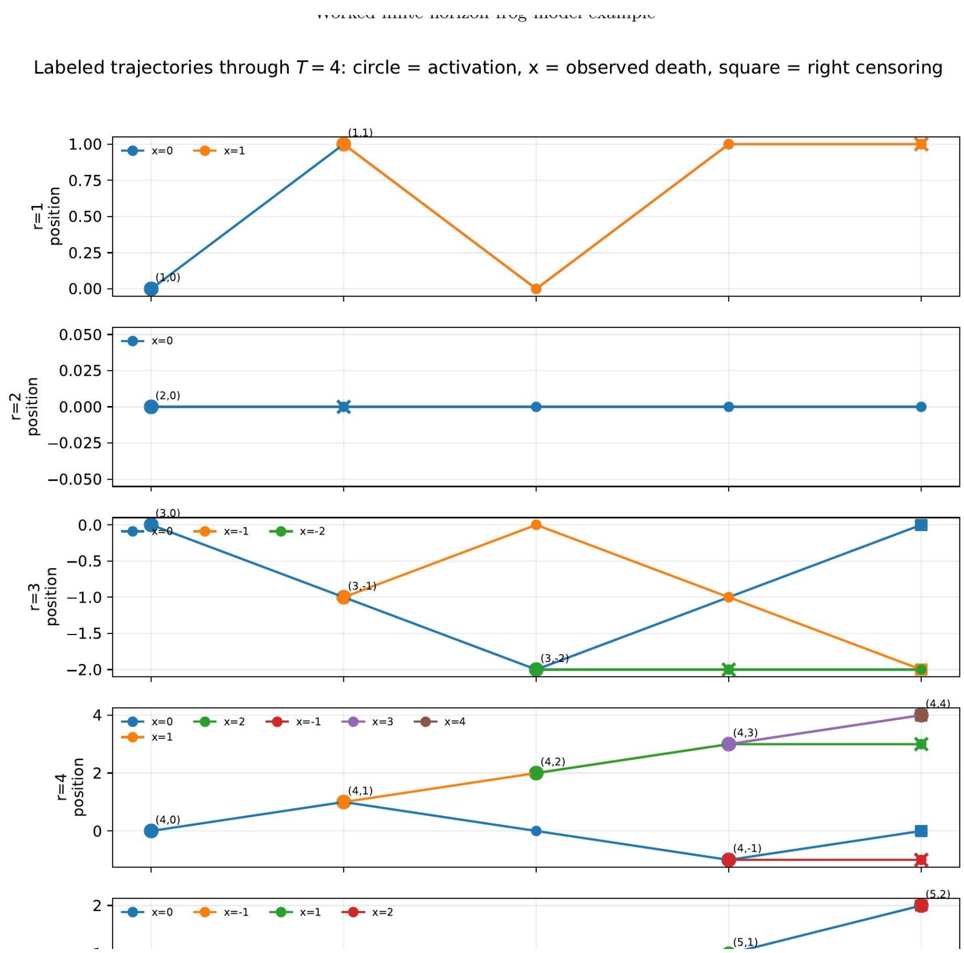

FIGURE 1. Labeled particle trajectories through the finite horizon.

jump, and is right-censored. Particle $(5, 2)$ is activated at time T and therefore contributes no transition of its own.

3.1. **Global activation order.** The sequence $Z_n^{(5)}$ begins with the five origin particles. The remaining particles are ordered by activation time, with deterministic tie-breaking. Figure 2 represents this order graphically, while Table 1 gives the corresponding latent values.

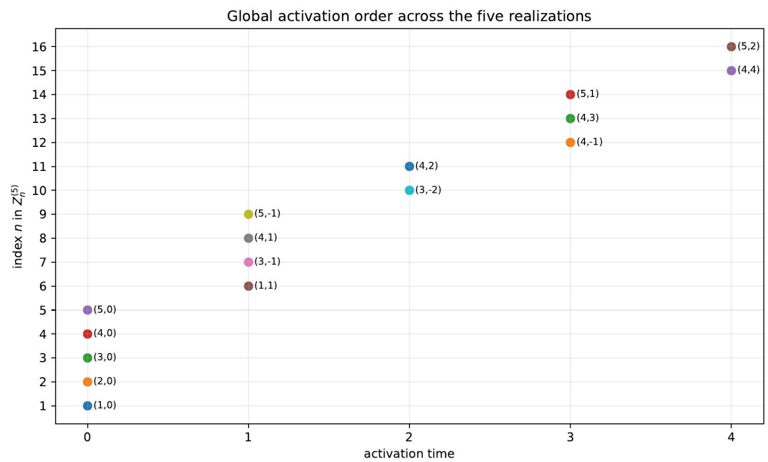

FIGURE 2. Global activation index n and activation time of $Z_n^{(5)}$.

TABLE 1. Activation order and associated latent values. Survival marks are displayed to twelve decimal places.

n	$Z_n^{(5)} = (r, x)$	$A_{r,x}$	$\Pi_{r,x}$	$\Xi_{r,x}$
1	$(1, 0)$	0	0.748852052343	3
2	$(2, 0)$	0	0.658864800511	0
3	$(3, 0)$	0	0.867178100529	14
4	$(4, 0)$	0	0.874058328918	14
5	$(5, 0)$	0	0.850211770952	24
6	$(1, 1)$	1	0.679706918303	2
7	$(3, -1)$	1	0.840255727892	11
8	$(4, 1)$	1	0.965480621624	42
9	$(5, -1)$	1	0.547795716564	0
10	$(3, -2)$	2	0.146829468108	0
11	$(4, 2)$	2	0.377623703115	1
12	$(4, -1)$	3	0.474265696617	0
13	$(4, 3)$	3	0.677549874181	4
14	$(5, 1)$	3	0.947083752203	28
15	$(4, 4)$	4	0.978962476308	0
16	$(5, 2)$	4	0.995044411438	56

4. PARTICLE-LEVEL REPRESENTATION OF THE RECORD

Table 2 collects the principal latent and observed variables. The column $X(0{:}Z)$ gives the observed segment of the potential path. A particle activated at T has $C = Z = 0$, $\Delta = 0$, and contributes $S_\beta(0) = 1$ through its own lifetime.

Figure 3 isolates the distinction between an observed death and right censoring. A cross marks death within the observation window; a square marks termination of observation while the particle remains alive. Particles activated at T have zero exposure after activation.

TABLE 2. Latent and observed values for every activated particle.

n	(r,x)	A	Π	Ξ	C	Z	Δ	$X(0{:}Z)$	status at T
1	$(1,0)$	0	0.748852052343	3	4	3	1	$(0,1,0,1)$	dead
2	$(2,0)$	0	0.658864800511	0	4	0	1	(0)	dead
3	$(3,0)$	0	0.867178100529	14	4	4	0	$(0,-1,-2,-1,0)$	censored
4	$(4,0)$	0	0.874058328918	14	4	4	0	$(0,1,0,-1,0)$	censored
5	$(5,0)$	0	0.850211770952	24	4	4	0	$(0,-1,0,1,2)$	censored
6	$(1,1)$	1	0.679706918303	2	3	2	1	$(1,0,1)$	dead
7	$(3,-1)$	1	0.840255727892	11	3	3	0	$(-1,0,-1,-2)$	censored
8	$(4,1)$	1	0.965480621624	42	3	3	0	$(1,2,3,4)$	censored
9	$(5,-1)$	1	0.547795716564	0	3	0	1	(-1)	dead
10	$(3,-2)$	2	0.146829468108	0	2	0	1	(-2)	dead
11	$(4,2)$	2	0.377623703115	1	2	1	1	$(2,3)$	dead
12	$(4,-1)$	3	0.474265696617	0	1	0	1	(-1)	dead
13	$(4,3)$	3	0.677549874181	4	1	1	0	$(3,4)$	censored
14	$(5,1)$	3	0.947083752203	28	1	1	0	$(1,0)$	censored
15	$(4,4)$	4	0.978962476308	0	0	0	0	(4)	active at T
16	$(5,2)$	4	0.995044411438	56	0	0	0	(2)	active at T

article remains alive. Particles activated exactly at T have zero exposure after activation.

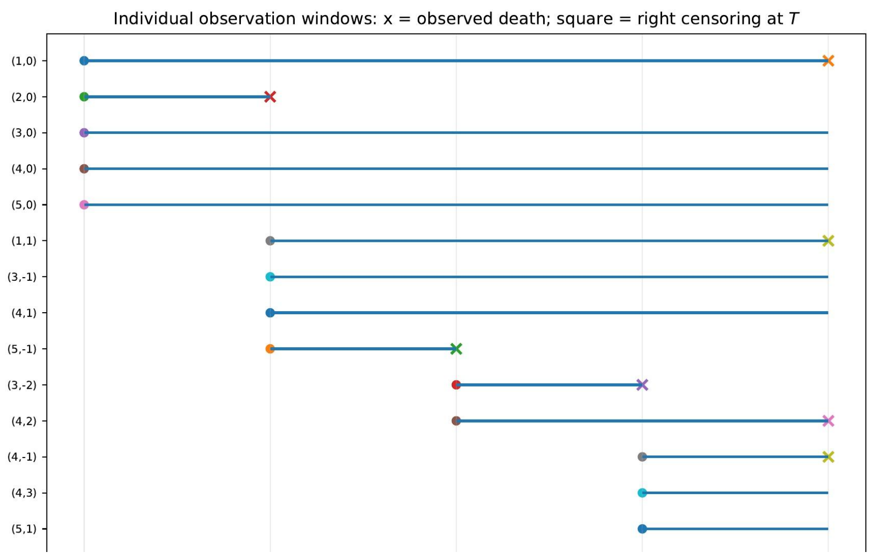

FIGURE 3. Individual observation windows and terminal observation type.

4.1. **Observed state–position record.** In Table 3, $\mathrm{sl}(x)$ denotes a dormant particle at its home site, $\mathrm{al}(y)$ an active and alive particle at position y, and $\mathrm{de}(y)$ a dead particle whose last occupied position is retained for bookkeeping.

TABLE 3. Observed values of $(H_{r,x}(t), L_{r,x}(t))$ at each calendar time.

Particle	$t=0$	$t=1$	$t=2$	$t=3$	$t=4$
$(1,0)$	al(0)	al(1)	al(0)	al(1)	de(1)
$(2,0)$	al(0)	de(0)	de(0)	de(0)	de(0)
$(3,0)$	al(0)	al(−1)	al(−2)	al(−1)	al(0)
$(4,0)$	al(0)	al(1)	al(0)	al(−1)	al(0)
$(5,0)$	al(0)	al(−1)	al(0)	al(1)	al(2)
$(1,1)$	sl(1)	al(1)	al(0)	al(1)	de(1)
$(3,-1)$	sl(−1)	al(−1)	al(0)	al(−1)	al(−2)
$(4,1)$	sl(1)	al(1)	al(2)	al(3)	al(4)
$(5,-1)$	sl(−1)	al(−1)	de(−1)	de(−1)	de(−1)
$(3,-2)$	sl(−2)	sl(−2)	al(−2)	de(−2)	de(−2)
$(4,2)$	sl(2)	sl(2)	al(2)	al(3)	de(3)
$(4,-1)$	sl(−1)	sl(−1)	sl(−1)	al(−1)	de(−1)
$(4,3)$	sl(3)	sl(3)	sl(3)	al(3)	al(4)
$(5,1)$	sl(1)	sl(1)	sl(1)	al(1)	al(0)
$(4,4)$	sl(4)	sl(4)	sl(4)	sl(4)	al(4)
$(5,2)$	sl(2)	sl(2)	sl(2)	sl(2)	al(2)

4.2. **Particle–age rows.** For each activated particle, Table 4 reports the vectors (Y_0, Y_1, Y_2, Y_3), (D_0, D_1, D_2, D_3), and (J_0, J_1, J_2, J_3). These are the indicators entering the hazard representation of the likelihood.

TABLE 4. At-risk, death, and jump indicators by individual age.

Particle	A	(Y_0, Y_1, Y_2, Y_3)	(D_0, D_1, D_2, D_3)	(J_0, J_1, J_2, J_3)
$(1,0)$	0	$(1,1,1,1)$	$(0,0,0,1)$	$(1,1,1,0)$
$(2,0)$	0	$(1,0,0,0)$	$(1,0,0,0)$	$(0,0,0,0)$
$(3,0)$	0	$(1,1,1,1)$	$(0,0,0,0)$	$(1,1,1,1)$
$(4,0)$	0	$(1,1,1,1)$	$(0,0,0,0)$	$(1,1,1,1)$
$(5,0)$	0	$(1,1,1,1)$	$(0,0,0,0)$	$(1,1,1,1)$
$(1,1)$	1	$(1,1,1,0)$	$(0,0,1,0)$	$(1,1,0,0)$
$(3,-1)$	1	$(1,1,1,0)$	$(0,0,0,0)$	$(1,1,1,0)$
$(4,1)$	1	$(1,1,1,0)$	$(0,0,0,0)$	$(1,1,1,0)$
$(5,-1)$	1	$(1,0,0,0)$	$(1,0,0,0)$	$(0,0,0,0)$
$(3,-2)$	2	$(1,0,0,0)$	$(1,0,0,0)$	$(0,0,0,0)$
$(4,2)$	2	$(1,1,0,0)$	$(0,1,0,0)$	$(1,0,0,0)$
$(4,-1)$	3	$(1,0,0,0)$	$(1,0,0,0)$	$(0,0,0,0)$
$(4,3)$	3	$(1,0,0,0)$	$(0,0,0,0)$	$(1,0,0,0)$
$(5,1)$	3	$(1,0,0,0)$	$(0,0,0,0)$	$(1,0,0,0)$
$(4,4)$	4	$(0,0,0,0)$	$(0,0,0,0)$	$(0,0,0,0)$
$(5,2)$	4	$(0,0,0,0)$	$(0,0,0,0)$	$(0,0,0,0)$

Summing over particles gives

$$(R(0), R(1), R(2), R(3)) = (14, 8, 7, 4), \qquad (d(0), d(1), d(2), d(3)) = (4, 1, 1, 1).$$

Hence the visible-jump counts by age are $(10, 7, 6, 3)$, and the total number of observed jumps is

$$G_{5,4} = 10 + 7 + 6 + 3 = 26.$$

$(0), R(1), R(2), R(3)) = (14, 8, 7, 4), \qquad (d(0), d(1), d(2), d(3)) = (4, 1, 1, 1).$

s of visible jumps by age are therefore $(10, 7, 6, 3)$, and the total number of

$$G_{5,4} = 10 + 7 + 6 + 3 = 26.$$

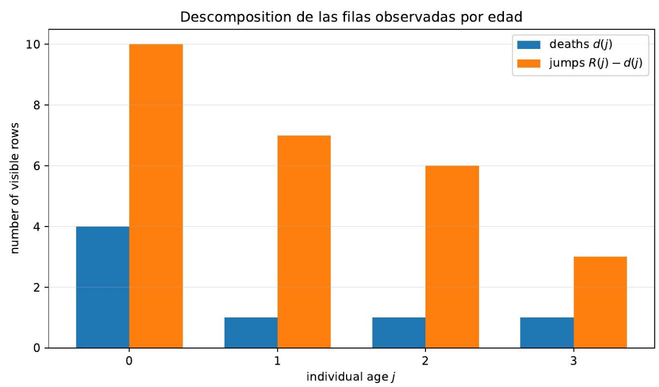

FIGURE 4. Observed deaths and jumps by individual age.

5. First-transition estimators

5.1. Estimator based on the origin particles. Let

$$I_r = \mathbf{1}_{\{\Xi_{r,0} \geq 1\}}.$$

For the present record,

$$(\Xi_{1,0}, \ldots, \Xi_{5,0}) = (3, 0, 14, 14, 24), \qquad (I_1, \ldots, I_5) = (1, 0, 1, 1, 1).$$

Therefore,

$$\widehat{q}_{\mathrm{root}} = \frac{4}{5}, \qquad \widehat{\beta}_{\mathrm{root}} = \widehat{q}_{\mathrm{root}}^{-1} - 1 = \frac{1}{4} = 0.25.$$

This estimator uses the guaranteed first transition of the origin particle in each realization.

5.2. Pooled estimator based on visible first transitions. Exactly

$$N_{5,4} = 14$$

particles are activated strictly before the horizon and therefore contribute visible first transitions. Of these, $M_{5,4} = 10$ complete a jump. Thus

$$\widehat{q}_{\mathrm{act}} = \frac{10}{14} = \frac{5}{7}, \qquad \widehat{\beta}_{\mathrm{act}} = \widehat{q}_{\mathrm{act}}^{-1} - 1 = \frac{2}{5} = 0.4.$$

Although this ratio has the form of a binomial estimator, its justification is not a conditional binomial law for $M_{5,4}$ given $N_{5,4}$. The number and identities of the contributing particles are generated by the process; the corresponding argument uses predictable revelation and the associated martingale estimating identity.

6. Likelihood evaluation for the observed record

The survival function, age-specific death hazard, and lifetime mass function are

$$S_\beta(j) = \mathbb{P}_\beta(\Xi \geq j), \qquad h_\beta(j) = \mathbb{P}_\beta(\Xi = j \mid \Xi \geq j) = \frac{\beta}{\beta + j + 1}, \qquad p_\beta(j) = S_\beta(j) h_\beta(j).$$

6.1. **Particle-level censored-lifetime representation.** Observed deaths occur for

$$(1,0),(2,0),(5,-1),(3,-2),(4,-1),(4,2),(1,1),$$

with observed lifetimes $3,0,0,0,0,0,1,2$, respectively. The right-censored particles activated before T contribute three factors $S_\beta(4)$, two factors $S_\beta(3)$, and two factors $S_\beta(1)$. The two particles activated at T contribute $S_\beta(0)=1$. Consequently,

$$\boxed{L_{5,4}(\beta)=2^{-26}p_\beta(0)^4 p_\beta(1)p_\beta(2)p_\beta(3)S_\beta(4)^3 S_\beta(3)^2 S_\beta(1)^2.} \tag{1}$$

The factor 2^{-26} is the probability of the 26 observed directions. No additional activation factor is needed: conditional on the preceding transitions and directions, the activations in the labeled record are determined.

6.2. **Age-compressed hazard representation.** Equivalently,

$$L_{5,4}(\beta)=2^{-26}\prod_{j=0}^{3} h_\beta(j)^{d(j)}\{1-h_\beta(j)\}^{R(j)-d(j)}.$$

Substitution of the observed counts yields

$$\boxed{\begin{aligned} L_{5,4}(\beta)=2^{-26}&\left(\frac{\beta}{\beta+1}\right)^4\left(\frac{1}{\beta+1}\right)^{10}\left(\frac{\beta}{\beta+2}\right)\left(\frac{2}{\beta+2}\right)^7\\ &\times\left(\frac{\beta}{\beta+3}\right)\left(\frac{3}{\beta+3}\right)^6\left(\frac{\beta}{\beta+4}\right)\left(\frac{4}{\beta+4}\right)^3. \end{aligned}} \tag{2}$$

This form displays the contribution of every visible death, jump, and terminal censoring event through the age-specific counts.

6.3. **Maximum-likelihood estimate.** Let $\theta=\log\beta$. Up to terms independent of θ, the log-likelihood is

$$\ell_{5,4}(\theta)=7\theta-14\log(e^\theta+1)-8\log(e^\theta+2)-7\log(e^\theta+3)-4\log(e^\theta+4).$$

Its score is

$$U_{5,4}(\theta)=7-14h_\theta(0)-8h_\theta(1)-7h_\theta(2)-4h_\theta(3).$$

Thus $U_{5,4}(\widehat{\theta})=0$ is equivalent to

$$7=\frac{14\beta}{\beta+1}+\frac{8\beta}{\beta+2}+\frac{7\beta}{\beta+3}+\frac{4\beta}{\beta+4},$$

whose positive solution is

$$\widehat{\beta}_{\mathrm{MLE}}\approx 0.4390.$$

Figure 5 shows the relative likelihood together with its maximizer and the generating value $\beta_0=0.45$.

solution is

$$\widehat{\beta}_{\mathrm{MLE}} \approx 0.4390.$$

plays the relative likelihood and compares its maximizer with the genera

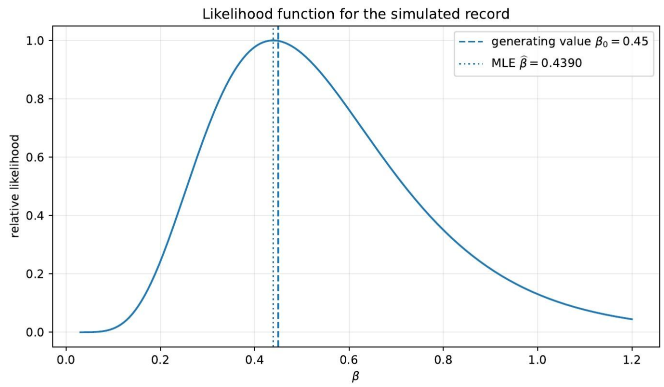

FIGURE 5. Relative likelihood for the simulated labeled record.

7. Numerical summary

For this record,

$$\beta_0 = 0.45, \qquad \widehat{\beta}_{\mathrm{root}} = 0.25, \qquad \widehat{\beta}_{\mathrm{act}} = 0.40, \qquad \widehat{\beta}_{\mathrm{MLE}} \approx 0.4390.$$

The origin estimator uses five binary observations, the pooled estimator uses 14 visible first transitions, and the maximum-likelihood estimator also uses the transitions at ages $1, 2, 3$ and terminal right censoring. In this realization, the latter estimate is closest to the generating value.

Because $\beta_0 = 0.45 < 1/2$, the generating value lies in the positive-survival regime of the off-critical phase result, and all three point estimates also fall below $1/2$. This comparison is descriptive only; formal phase classification requires a confidence set or a hypothesis test.

Remark 7.1. *The record also distinguishes right censoring from a death observed at the horizon. If a particle is alive at T, the data imply only $\Xi \geq C(T)$ and the likelihood contains $S_\beta(C(T))$. If the particle is recorded as dead at T, death occurred during the final observed transition and the value of Ξ is known, so the likelihood contains $p_\beta(\Xi)$. A particle activated at T has $C(T) = 0$ and contributes $S_\beta(0) = 1$ through its own lifetime; the jump that activated it is already represented in the preceding particle's observed transition.*

Reproducibility files

The accompanying repository directory contains this LaTeX source, the five figure files in PDF format, the simulation output in JSON format, and the Python script used to reproduce the record and figures.